

My engineering notebook

Engineers do not get it right first time. They go round a loop: ask what the problem really is, imagine several answers, plan one of them, build it, then find out what is wrong with it and go round again. This notebook is that loop, with room to write in.

THE THING I AM BUILDING

THIS NOTEBOOK BELONGS TO

STARTED ON

1

Ask

What is the problem, exactly?

Who is it for, what has to happen, and what counts as working? A problem you cannot say in one sentence is still two problems.

2

Imagine

What are all the ways I could do it?

Three ideas, not one. The first idea is rarely the best one, and you only find that out by having a second and a third to compare it with.

3

Plan

How will this one actually go together?

Draw it big. Label the parts. List what you need. A drawing is cheaper to change than a build.

4

Create

Does it work?

Build it and test it properly, the same way every time, so you can trust what the test tells you.

5

Improve

What broke, and what will I change?

Find the weakest part, change one thing, test again. This is the part real engineers spend most of their time on.

Then go round again. The loop has no finish line. A design is finished when you decide to stop improving it, and every real engineer stops somewhere.

1. Ask · 2. Imagine

The problem, in one sentence

Start with: I want to make something that...

It works if...

Three things you could actually check or measure.

Rules I have to obey

Materials I am allowed to use

How big it can be

How long I have

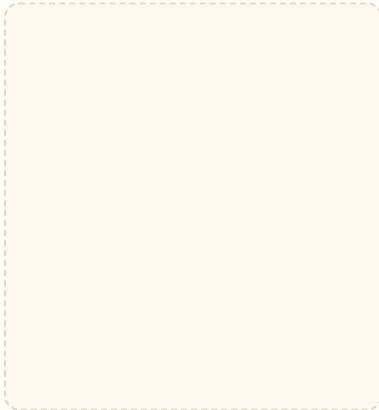
What must not happen (safety)

It is for

Three different ideas

Different, not three versions of the same thing. Under each one, write the best thing and the worst thing about it.

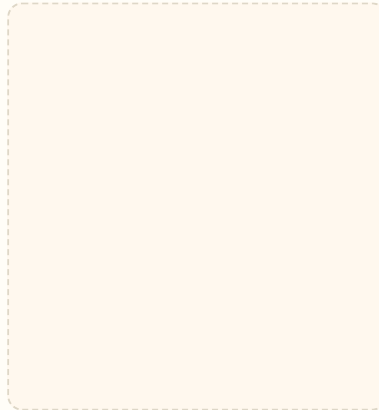
Idea 1



Best thing:

Worst thing:

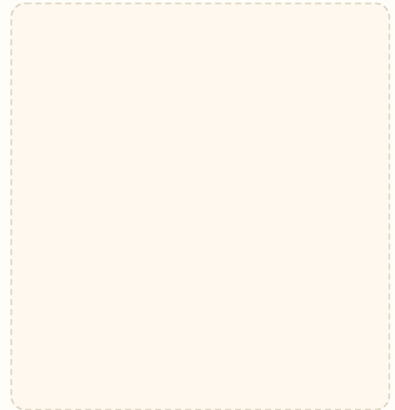
Idea 2



Best thing:

Worst thing:

Idea 3



Best thing:

Worst thing:

3. Plan

The idea I picked, and why

Draw it big and label every part

Label what each part does, not just what it is called. Mark the joints. Mark which way the forces push.

What I need

PART	HOW MANY	HAVE IT?

The part I am least sure about

Every design has one. Naming it before you build is what separates a plan from a hope.

The first thing I will test

4. Create · 5. Improve

Test the same way every round, change one thing at a time, and write it down before you forget it. A round where it broke teaches you more than a round where it held.

Round 1

WHAT I CHANGED

WHAT HAPPENED

WHAT I LEARNED

Round 2

WHAT I CHANGED

WHAT HAPPENED

WHAT I LEARNED

Round 3

WHAT I CHANGED

WHAT HAPPENED

WHAT I LEARNED

The version I would build next

The thing that surprised me most

The part I am proudest of

Nothing here is a mistake. A bridge that folds, a catapult that fires backwards, a tower that leans: each one just told you something true about your design that no amount of thinking would have. That is what a test is for.